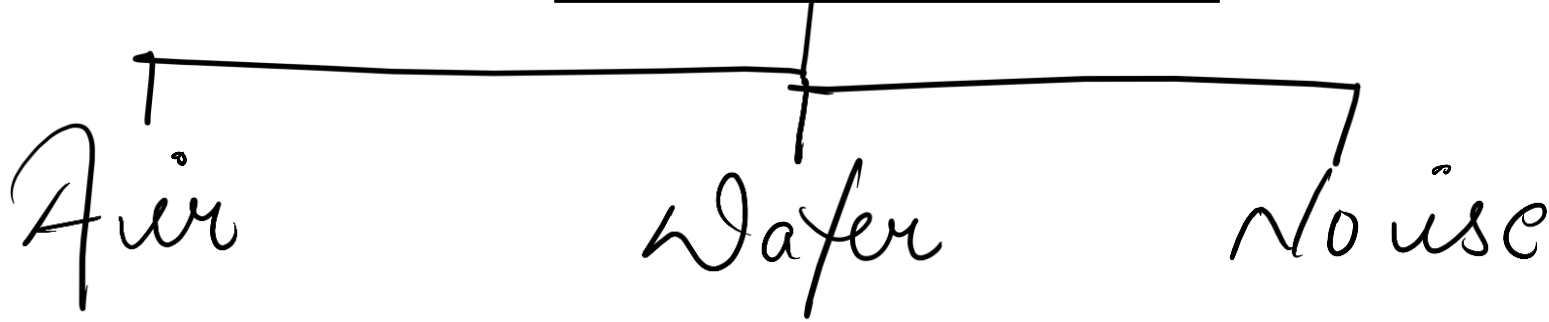


MINE ENVIRONMENT



① Water

→ waste water quality Parameters.

- ① Physical
- ② Chemical
- ③ Biological.

① Physical → Suspended particles (temperature, pressure, colour, taste, turbidity)

② Chemical → Total suspended solids (pH, N_2 , O_2 , F, metals etc).

③ Biological → Pathogens.

Water has a pH.

$$pH = -\log [H^+]$$

$$pOH = -\log [OH^-]$$

★★ $pH + pOH = 14$

$$[H^+][OH^-] = 10^{-14} \quad \text{--- ①}$$

$$pH = -\log[H^+], \quad pOH = -\log[OH^-]$$

✓ Take suppose,

$$pH = 7, \quad pOH = 14 - pH = 14 - 7 = 7$$

$$7 = -\log[H^+]$$

$$-7 = \log[H^+]$$

$$[H^+] = 10^{-7}$$

$$7 = -\log[OH^-]$$

$$-7 = \log[OH^-]$$

$$[OH^-] = 10^{-7}$$

$$[H^+]=10^{-7}$$

$$9, [\text{OH}^-] = 10^{-7}$$

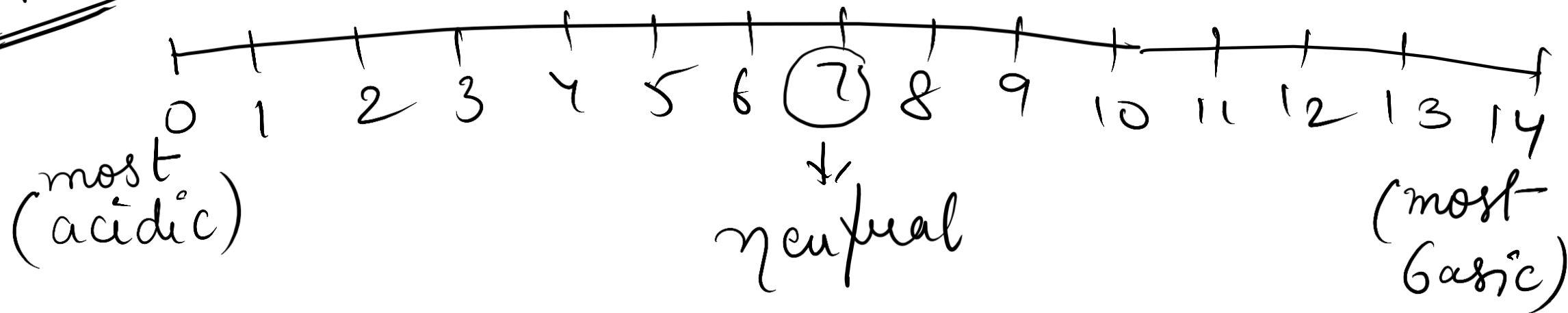
$$[H^+][OH^-] = 10^{-7} \times 10^{-7} = 10^{-14}$$

$$|P_H| = 7$$

(neural)

Prüfung

pH scale



(most acidic)

neutral

(most-basic)

$\text{pH} = 0$ } strong
 $\text{pOH} = 14$ } acid
Acidic

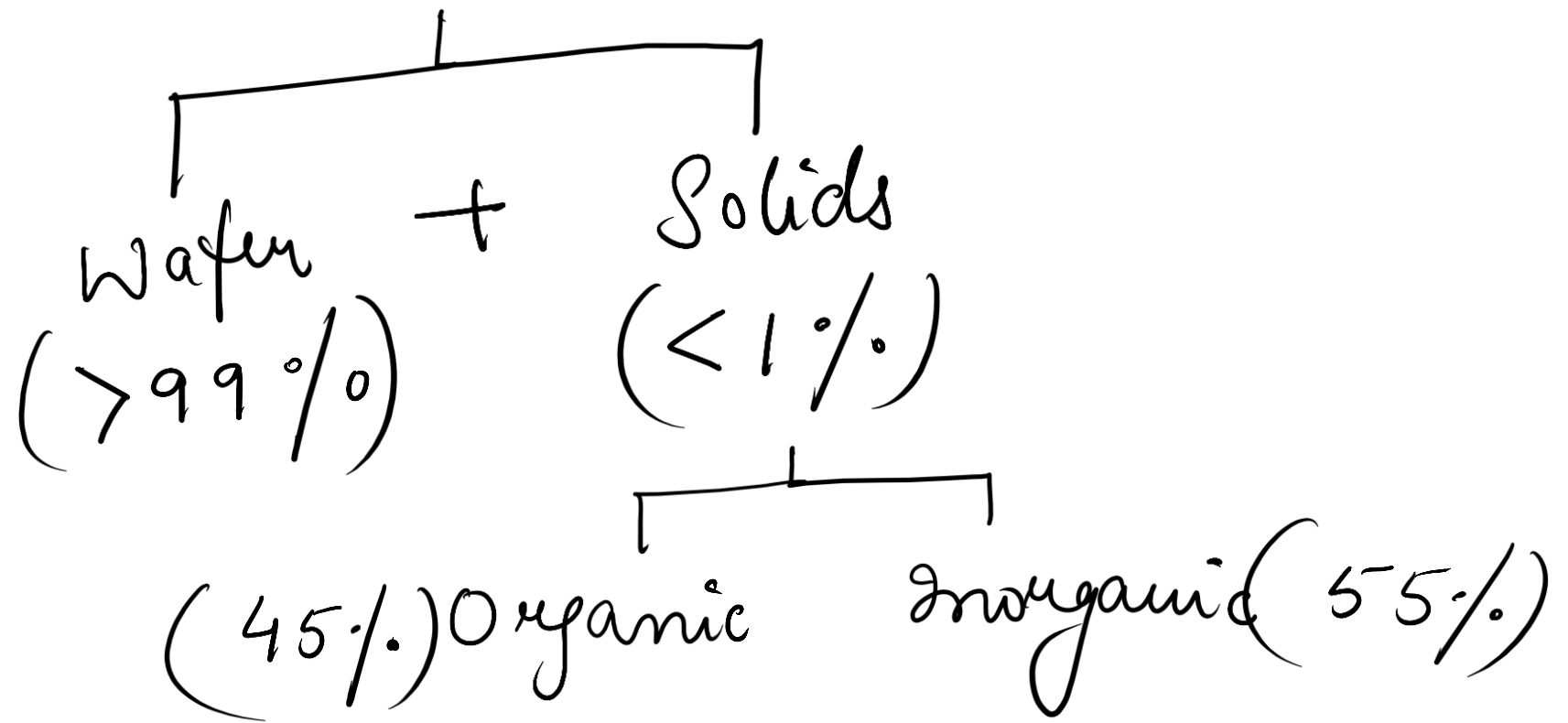
$\text{pH} = 14$ } strong
 $\text{pOH} = 0$ } base
Basic.

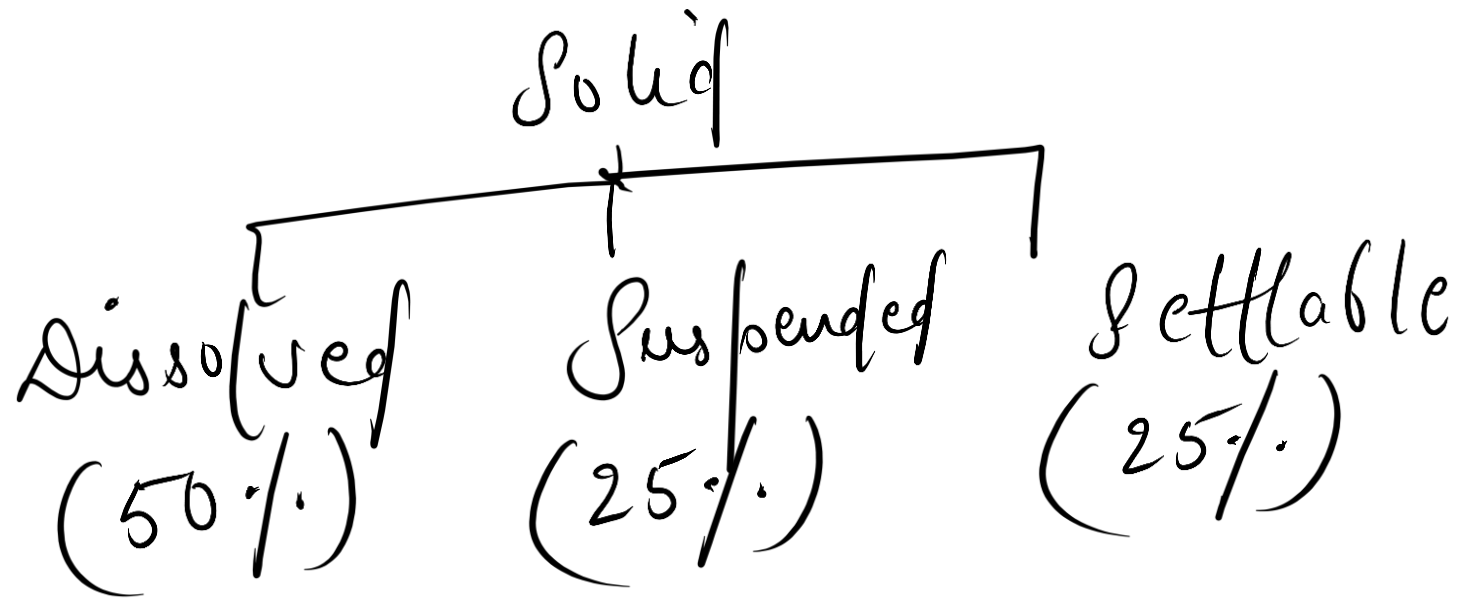
* Measurements.

- ① pH :- pH meter.
- ② colour indicator :- Phenolphthaleine, methyl orange.
- ③ Acceptable range $\rightarrow (7 - 8.5)$

Chemical Water Quality Parameters.

① Domestic
② Industrial
③ Rain water } → sewage.





Dissolved Oxygen (DO)

It is defined as the amount of oxygen present in it.

- Maximum amount of oxygen dissolved in water at a particular temp & pressure.
- Any deficiency in the level of DO indicates the presence of organic matter in it.

* Winkler's method is used to determine DO.

→ fish survival needs at least 4 ppm DO.

Biochemical Oxygen Demand (BOD)

→ it is defined as the measure of amount of DO needed by microorganisms to completely decompose the organic matter present within it.

VVI.

$$BOD_5 = \frac{(DO_i - DO_f) \times (\text{Waste water} + \text{Pure water})}{\text{Waste water}}$$